Reclaiming the GenAI Pipeline for EAP

Prompt Engineering, Data Dashboards, and Vibe Coding

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Overview

1. Introduction: Reclaiming the Pipeline
2. EAP for Prompt Engineering
3. Case Study – The Qualitative Research Dashboard
4. Vibe Coding in Practice
5. Implications & Next Steps

Introduction: Reclaiming the Pipeline

Focusing on Agency and Productivity

- **The Current Educational Discourse:**
  - Dominated by academic integrity (preventing cheating).
  - Focused on redesigning assessments to be “AI-proof.”
  - Overflow of similar GenAI-focused studies (Hu, 2026)
- **A Shift in Perspective:**
  - Refocus on personal productivity and linguistic agency.
  - Applied linguists and EAP scholars are uniquely positioned for this space. (Jiang and Hyland, 2026)
- **The Core Message (Affordances of AI):**
  - Prompt engineering is fundamentally about refining natural language input.
  - EAP practitioners understand writing, genre structure, and academic literacy.
  - We should claim expertise in prompting, using it for both teaching and research.

EAP for Prompt Engineering

Prompt Engineering Terminology

- **Prompt:** The natural language input and instructions passed to AI models.
- **Completion:** The output generated in response. It is a “completion” because the statistical engine predicts how to complete the sequence of text.
- **Model:** The underlying statistical engine trained on massive datasets to predict the most likely next word (or ‘token’) based on patterns.
- **Agent:** Entities in computer science designed to act with varying degrees of autonomy to achieve goals.

Mapping EAP Expertise to AI Prompting

Prompting Step	**For AI**	**EAP / Applied Linguistics Expertise**
Task	Defining the Action	**Genre Knowledge** (Rhetorical moves & conventions)
Context	Targeted Background	**Needs & Register Analysis** (Audience, discipline)
References	Provide desired output	**Selecting Authentic Exemplars** (Corpora & model texts)
Evaluate	Analyzing AI's output	**Critical Evaluation of Texts** (Cohesion, stance, voice)
Iterate	Refining the prompt	**Process Writing** (Drafting, feedback, editing)

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Writing prompts is writing instruction. EAP practitioners already have the frameworks.

Linguistic Prompting Techniques

- **Chain-of-Thought (CoT) Prompting:**
  - Breaking down complex tasks into logical, step-by-step instructions.
  - *Example:* “Evaluate this draft. First, identify transition words. Second, map how they connect ideas. Finally, suggest improvements for flow.”

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- **Few-Shot Prompting (Using References):**
  - Feeding high-quality academic exemplars directly into the model to set standards for tone, stance, and formatting (e.g. abstract grading rubrics).

Configuring Custom EAP Agents

- We can encapsulate EAP frameworks into custom assistants (e.g. MS Copilot custom agents, Google AI Studio templates).

Agent	**System Instruction**	**EAP Skill / Theoretical Core**
Hedging Detector	'Identify bald claims and suggest three levels of hedging (cautious, moderate, strong).'	**Pragmatic competence** (Authorial voice and stance)
Move-Structure Analyzer	'Using Swales' CARS model as a reference, identify rhetorical moves in this abstract.'	**Genre analysis** (Applying structural frameworks)
AWL Contextualizer	'Identify Academic Word List words. Provide domain-specific definition and example sentences.'	**Lexical range** (Vocabulary expansion in context)

Leveraging GenAI in Research / Project Management

- NOT writing!
- Think about the most repetitive and mechanical parts of data-driven research:
  - data collection (in corpus and experimental studies)
  - annotation
  - coding & formatting
  - visualization
  - figures and video stimuli (cf. data collection in experiments and elicitation)
- Note that we're talking about GenAI tools, not just chatbots!

Case Study – The Qualitative Research Dashboard

Pedagogical Gap in ESAP

- **The Context:** Medical, healthcare, and STEM students frequently write qualitative methods sections (dissertations/dissertation chapters) over the summer.
- **The Challenge:** Unfamiliarity with qualitative methodologies. Limited access to supervisors. High stakes, high anxiety.
- **Data-Driven Learning (DDL):**
  - Fosters lexico-grammatical awareness through exposure to authentic research articles.
  - STEM students are receptive to empirical evidence, but find general-purpose corpus interfaces (e.g., Sketch Engine) opaque and intimidating.
- **The Goal:** A bespoke, scaffolded digital tool designed specifically for healthcare qualitative writing, minimizing technical overhead.

Building the Custom Corpus

Data Collection & Curation

- Targeted research articles matching "qualitative study" in abstract or title.
- Isolated the methods section.
- Extracted via NCBI E-utilities API (Sayers, 2022)

The Datasets

- **PubMed Central:**
  - 150 articles
  - 111,043 tokens
  - 740 tokens average per section
  - full corpus: 15,159 articles with ca. 1.4 million tokens
- **PLoS One:**
  - 1,612 articles
  - 1,993,823 tokens
  - 1,236 tokens average per section

Dashboard and Key Features

`https://qual-dashboard.streamlit.app/`

- **Corpus Analysis:** Explores structural stats (e.g. section length) to answer: “How much do I need to write?”
- **Keyword & Concordance (KWIC) Search:** Locates disciplinary conventions (e.g., where and how ethical approvals are reported).
- **LDA Key Themes:** Automatic topic modeling (Latent Dirichlet Allocation) showing typical thematic clusters across 15,000+ publications.
- **Geographical Analysis:** Automatically maps research study locations using Named-Entity Recognition (NER).
- **Move-Step Analysis:** Rhetorical progression visualization mapping methods sections to the DRaC model (Cotos et al., 2017)

Development Process & Experience

- Started in March 2025 with Replit https://www.replit.com/ and Google AI Studio https://www.aistudio.google.com/ (not available in HK as of 10 July 2026), then Google Anitgravity https://antigravity.google.com/ (not available in HK as of 10 July 2026, but check out Cursor https://cursor.com/)
- First prototype in 30 minutes, then gradually added functions
- Iterative editing for each function and tab

Vibe Coding in Practice

What is Vibe Coding?

1. Describe what you want in plain English (or others languages really)
2. AI generates the code
3. Human users refine through conversation **iteratively**
   (See Andrej Karpathy’s X post)

Background of the Vibe Coding for Language Project

or: **Why stop at one webapp?**

- Feedback on the data dashboard: “But I don’t teach writing to biology MSc students!”
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Programming is no longer scary

- Programming syntax barrier is drastically reduced.
  But: syntax is not the only thing in programming language!
- Logical design, structuring information, and prompt precision are the new constraints.
- Language experts are already trained in these exact structural skills.

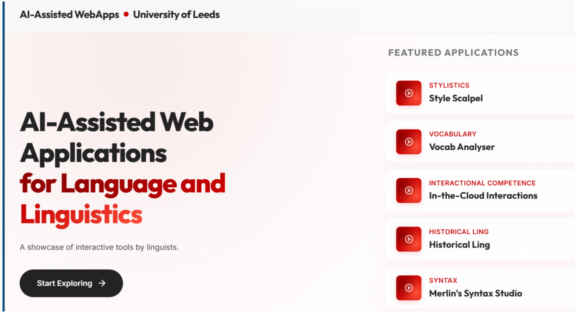
Output: Vibe Coding for Language Visualization Workshop

- Jan–May 2026
- Internally funded by Digital Cultures and Creativity Hub (DCCH) at Leeds
- 3 workshops and Replit subscriptions for participants
- Individual consultations

Vibe Coding Workshop & Showcase

A collection of AI-assisted tools by and for linguists:

`https://vibecoding4language.github.io/`



Custom Interactive Webapps

- **Pedagogical needs:**
  Tools for productivity and browsing through data
- **Implement via vibe coding:**
  Full control over how much / little to show; step-by-step guidance for users

GenAI for Research and more

Emerging **empirical** literature on using AI-assistants in linguistics research

- Using GenAI beyond teaching to directly assist with corpus annotation
  E.g. Move-step analysis (Yu et al., 2026a,b; Omotola et al., 2025)
- Both qualitative and quantitative studies (Crosthwaite, 2026)

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From the Vibe Coding Collection:

- Data visualization (Vocab Analyzer; DrameScope)
- Data collection (AI interlocutor; Style Scalpel)
- Displaying collection of work (Learning Gallery)
- Productivity (Merlin's Syntax Studio)

Implications & Next Steps

Implications for the EAP Practitioners

- **Empowerment through Technology**
  - Fast prototyping of dashboards can happen in minutes – less procrastination; more trial and error
  - Iterative process (build one part / one feature at a time – much like writing)

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- **The “Tech Stack”**
  - **Coding** / **Prototyping:** Google AI Studio, Replit, or local agentic environments (Google Antigravity SDK).
  - **Hosting** / **Sharing:** Streamlit Community Cloud (free deployment).

Conclusion

1. GenAI removes tech barriers for **Data-Driven Learning**.
2. **Prompt engineering and vibe coding** allow EAP practitioners to design these interfaces independently.
3. **Reclaiming educational technology** by guiding development using our domain expertise in linguistics, genre, and rhetorics.

Thank You! Questions?

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